

# FARM MANAGEMENT SYSTEM

## User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

## 1. INTRODUCTION

### 1.1 Project Overview:

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

### 1.2 Purpose:

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

## 2. IDEATION PHASE

### 2.1 Problem Statement (Customer Problem Statement Template)

| PS   | I am<br>(Customer) | I'm trying<br>to                       | But                                   | Because                                   | Which<br>makes me<br>feel  |
|------|--------------------|----------------------------------------|---------------------------------------|-------------------------------------------|----------------------------|
| PS-1 | Farmer             | Maintain farm and crop records         | Data is often incomplete or incorrect | Records are kept manually in registers    | Frustrated and confused    |
| PS-2 | Farm Manager       | Monitor farm productivity and sales    | Reports are delayed and inaccurate    | Data is scattered across files and sheets | Stressed and inefficient   |
| PS-3 | Buyer              | Access crop availability and pricing   | Information is not updated regularly  | There is no centralized digital system    | Uncertain and dissatisfied |
| PS-4 | System Admin       | Manage user access and system security | Unauthorized access may occur         | Access control is handled manually        | Concerned and insecure     |

### 2.2.2 Empathy Map Canvas

#### Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

#### Empathy Analysis

- **Think & Feel:** Need accurate and timely farm data
- **Hear:** Feedback from agricultural officers
- **See:** Registers, paper files, spreadsheets

- **Say & Do:** Enter records manually
- **Pain:** Dataloss, errors, delays
- **Gain:** Automation, fast reporting



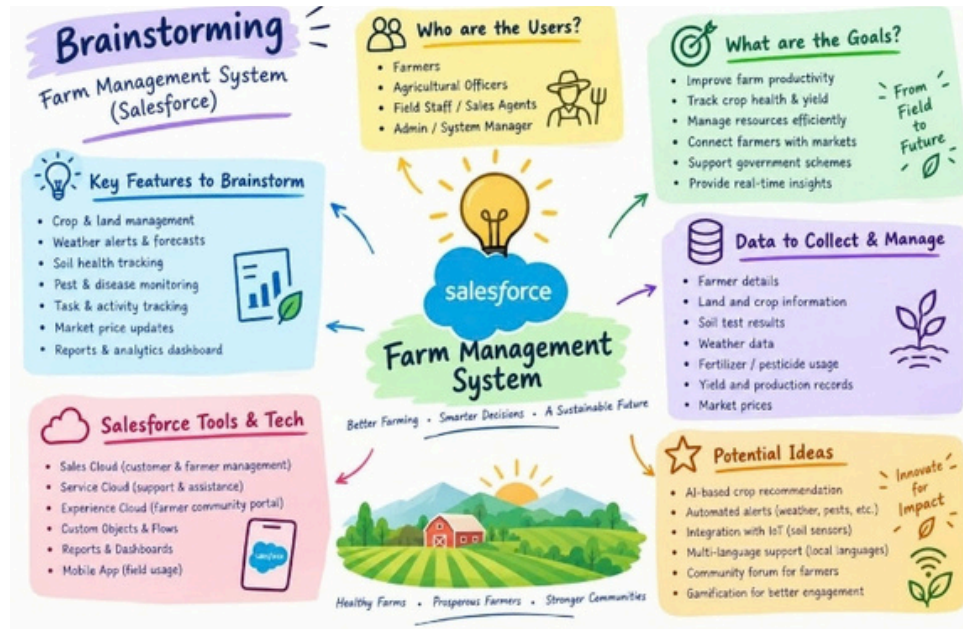
## 2.3 Brainstorming

### Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

### Selected Idea

Salesforce-based Farm Management System.



### 3.REQUIREMENT ANALYSIS PHASE

#### 3.1 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making



## 3.2 Solution Requirements

### Functional Requirements

| FR No | Requirement                          |
|-------|--------------------------------------|
|       |                                      |
| FR-1  | User Authentication                  |
| FR-2  | Farm, Crop, Farmer, Buyer Management |
| FR-3  | Data Entry and Validation            |
| FR-4  | Relationship Mapping                 |

|      |                         |
|------|-------------------------|
| FR-5 | Report Generation       |
| FR-6 | Dashboard Visualization |
| FR-7 | User Role Management    |

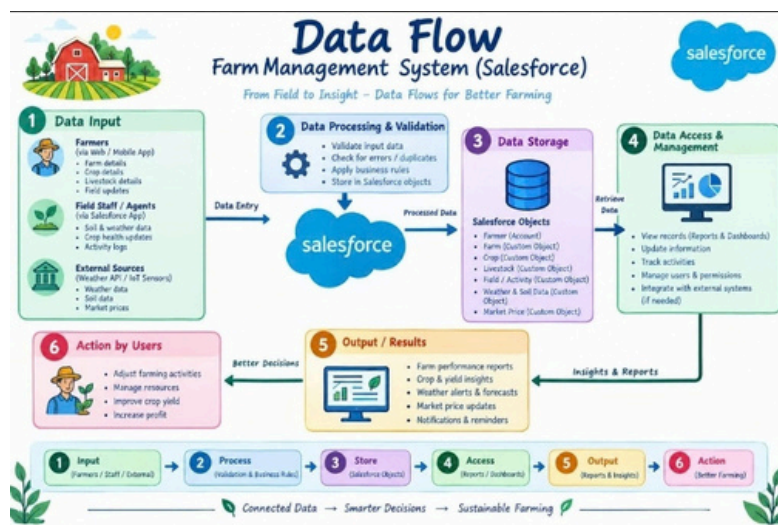
## Non-Functional Requirements

| NFR          | Description     | Example             |
|--------------|-----------------|---------------------|
| Usability    | Easy interface  | Simple Lightning UI |
| Security     | Data protection | Profiles & roles    |
| Reliability  | No data loss    | Cloud backup        |
| Performance  | Fast access     | Reports in <3 sec   |
| Availability | 24/7 access     | Cloud hosting       |
| Scalability  | Handles growth  | More farms/users    |

## 3.3 Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



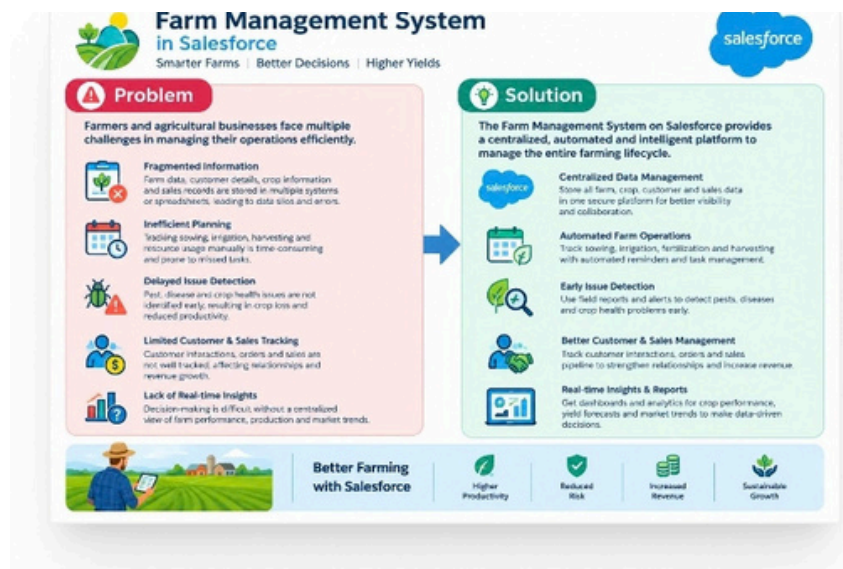
### 3.4 Technology Stack

| Layer      | Technology              |
|------------|-------------------------|
| Frontend   | Salesforce Lightning UI |
| Backend    | Salesforce Platform     |
| Database   | Salesforce Cloud        |
| Automation | Validation Rules        |
| Analytics  | Reports & Dashboards    |

## 4. PROJECT DESIGN PHASE

### 4.1 Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.



### 4.2 Proposed Solution

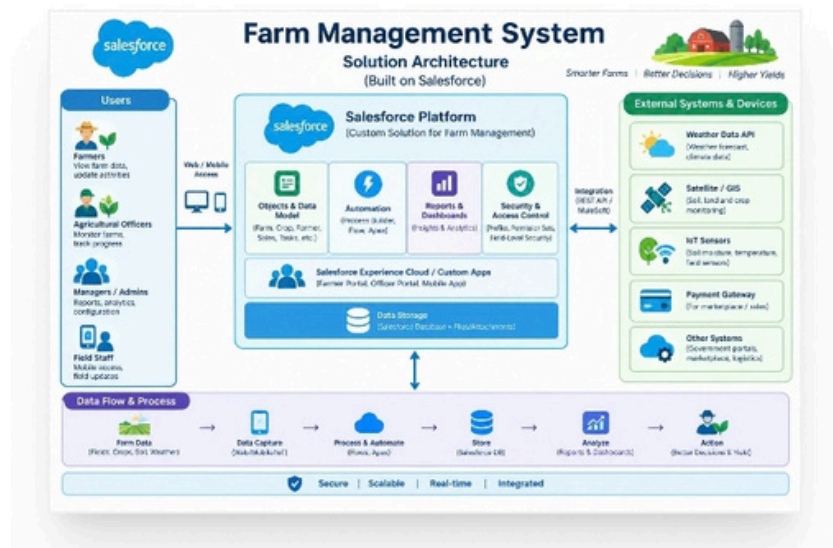
## Functional Requirements

| Parameter      | Description                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------|
| Problem        | Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays. |
| Solution       | Salesforce-based Farm Management System with centralized data storage and automation.                 |
| Innovation     | Automated data entry validation, cloud storage, and real-time reporting dashboards.                   |
| Impact         | Improved data accuracy, faster reporting, and better decision-making for farmers and managers.        |
| Business Model | Digital farm management and agricultural data services platform.                                      |
| Scalability    | Supports multiple farms, users, and regions with cloud-based infrastructure.                          |

## 4.3 Solution Architecture

### Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards



## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

| Sprint    | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                                                                    | Story Points | Priority | Team Members |
|-----------|-------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|--------------|
| Sprint-1  | Developer Setup               | USN-1             | As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System. | 3            | High     | Member1      |
| Sprint-2a | Data Modeling                 | USN-2             | As an admin, I want to create Farm, Crop, Farmer, and Buyer                                                                                          | 5            | High     | Member2      |

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                                                                            | Story Points | Priority | Team Members |
|----------|-------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|--------------|
|          |                               |                   | objects with proper relationships so that agricultural data is stored and managed efficiently.                                                               |              |          |              |
| Sprint-2 | Data Modeling                 | USN-3             | As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.                          | 5            | High     | Member3      |
| Sprint-3 | Automation                    | USN-4             | As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented. | 3            | High     | Member4      |
| Sprint-3 | Automation                    | USN-5             | As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.      | 3            | High     | Member4      |
| Sprint-4 | Security                      | USN-6             | As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.               | 5            | High     | Member2      |
| Sprint-5 | Reports                       | USN-7             | As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.      | 4            | High     | Member3      |
| Sprint-6 | Dashboards                    | USN-8             | As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.         | 4            | Medium   | Member4      |

#### Project Tracker, Velocity & Burndown Chart: (4 Marks)

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|----------|--------------------|----------|-------------------|---------------------------|-------------------------------------------------|------------------------------|
| Sprint-1 | 20                 | 2 Days   | 19 Sep 2026       | 20 Sept 2026              | 08                                              | 17 Sep 2026                  |
| Sprint-2 | 20                 | 2 Days   | 21 Sep 2026       | 22 Sep 2026               |                                                 |                              |
| Sprint-3 | 20                 | 2 Days   | 23 Sep 2026       | 24 Sep 2026               |                                                 |                              |
| Sprint-4 | 20                 | 2 Days   | 25 Sep 2026       | 26 Sep 2026               |                                                 |                              |

## **6. Project Development Phase:**

### **Project Description:**

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

### **Milestone 1-Salesforce:**

#### **What Is Salesforce?**

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

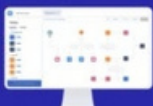
Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

#### **Activity1:**

##### **Creating a Salesforce Developer Org:**

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup)
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search [login.salesforce.com](https://login.salesforce.com)



### Build enterprise-quality apps fast to bring your ideas to life

Build apps fast with drag-and-drop tools  
Customize your data model with clicks  
Go further with Apex code  
Integrate with everything using powerful APIs  
Stay protected with enterprise-grade security  
Customize UI with clicks or any leading-edge web framework



### Sign up for your Salesforce Developer Edition

A full-featured copy of the Platform, for free

Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name\*

Your first name

Last Name\*

Your last name

Email\*

Your email address

Role\*

Your job role

Company\*

Company Name

Country/Region\*

Postal Code\*

Your postal code

Username\*

john@company.sandbox

Your username must be in the form of an email address (it does not have to be real). It must be unique and cannot be associated with another Salesforce login credential. Read more about username recommendations.

☐ I agree to the [Main Services Agreement](#) + [Developer Services and Salesforce Program Agreement](#).

Sign me up

6. By using username and password you can into the Salesforce org. The

homepage will appear below

Search Setup

Setup

Home

Object Manager

Quick Find

Setup Home

Service Setup Assistant

Commerce Setup Assistant

Hyperforce Assistant

Release Updates

Lightning Experience Transition Assistant

Salesforce Mobile App

Lightning Usage

Optimizer

Sales Cloud Everywhere

ADMINISTRATION

> Users

> Data

> Email

PLATFORM TOOLS

> Subscription Management

> Apps

> Feature Settings

> Slack

> Heroku

> Salesforce

SETUP

Home

Create

**Get Started with Einstein Bots**  
Launch an AI-powered bot to automate your digital connections.  
[Get Started](#)

**Mobile Publisher**  
Use the Mobile Publisher to create your web-branded mobile app.  
[Learn More](#)

**Real-time Collaborative Docs**  
Transform productivity with collaborative docs, spreadsheets, and slides inside Salesforce.  
[Get Started](#)

**Most Recently Used**

10 Items

| NAME                | TYPE                    | OBJECT  |
|---------------------|-------------------------|---------|
| SLA Serial Number   | Custom Field Definition | Account |
| Upsell Opportunity  | Custom Field Definition | Account |
| Action              | Custom Field Definition | Account |
| SLA Expiration Date | Custom Field Definition | Account |

## Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

**In This Application We Use 4 Custom Objects:**

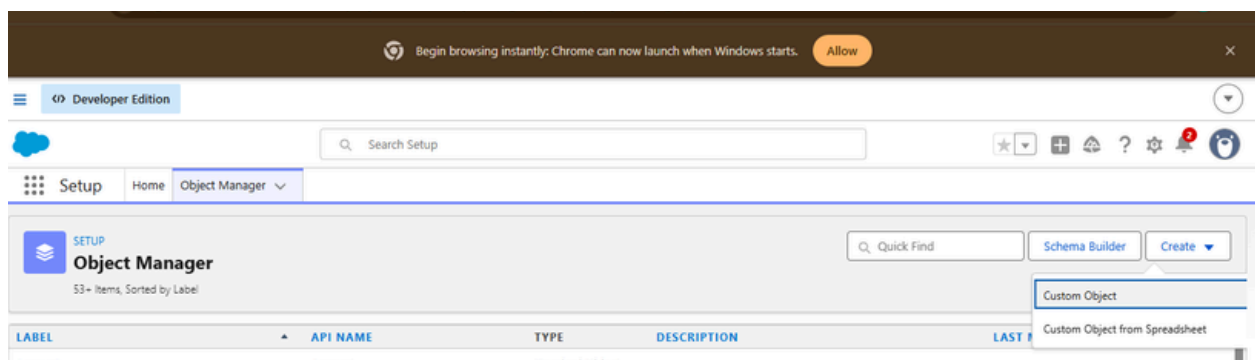
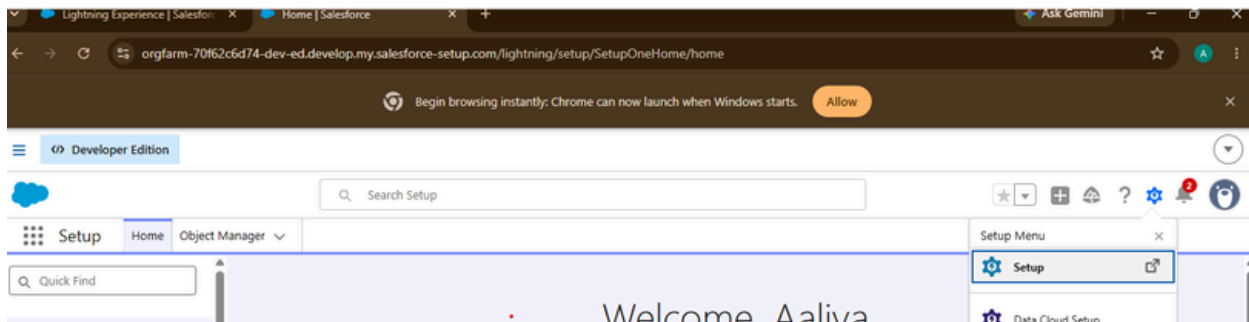
- 1.Farm
- 2.Crop
- 3.Farmer
- 4.Buyer

### Activity 1:

Create a custom object for Farm:

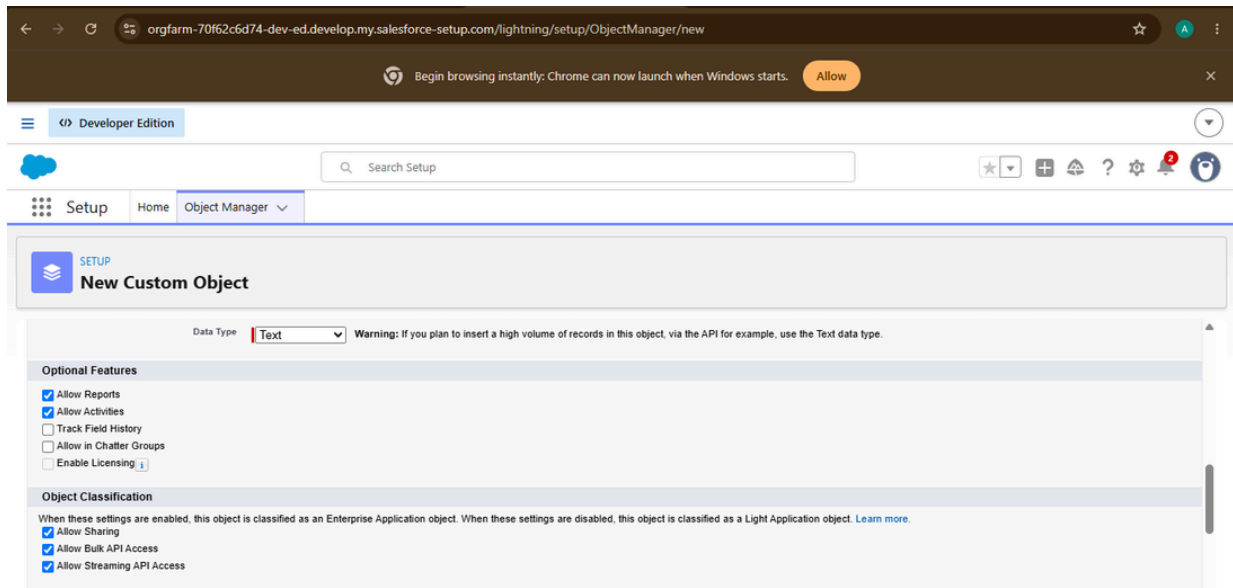
To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"

6. Select the data type as "Text".



7. In the Optional Features section, select Allow Reports and Track Field History.

8. In the Deployment Status section, ensure Deployed is selected.

9. In the Search Status section, select Allow Search.

10. In the Object Creation Options section, select Add Notes and Attachments related list

to default page layout.

11. Leave everything else as is, and click Save.

## **Activity 2:**

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
  - a. On the Custom Object Definition page, create the object as follows:
    - b. Label: Crop
    - c. Plural Label: Crops
    - d. Record Name: Crop Name
  - e. Select the data type as "Text".
  - f. Check the Allow Reports checkbox g. Check the Allow Search checkbox
  - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout i. Click Save.

## **Activity 3:**

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
  - a. On the Custom Object Definition page, create the object as follows:
    - b. Label: Farmer
    - c. Plural Label: Farmers
    - d. Record Name: Farmer Name e. Select the data type as "Text".
    - f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
    - h. In the Object Creation Options section, select Add Notes and Attachments related list

to default page layout. i. Click Save.

#### **Activity 4:**

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
- a. On the Custom Object Definition page, create the object as follows:
- b. Label: Buyer
- c. Plural Label: Buyers
- d. Record Name: Buyer Name
- e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
- i. Click Save.

#### **Milestone 3 -**

##### **Tab: Tab**

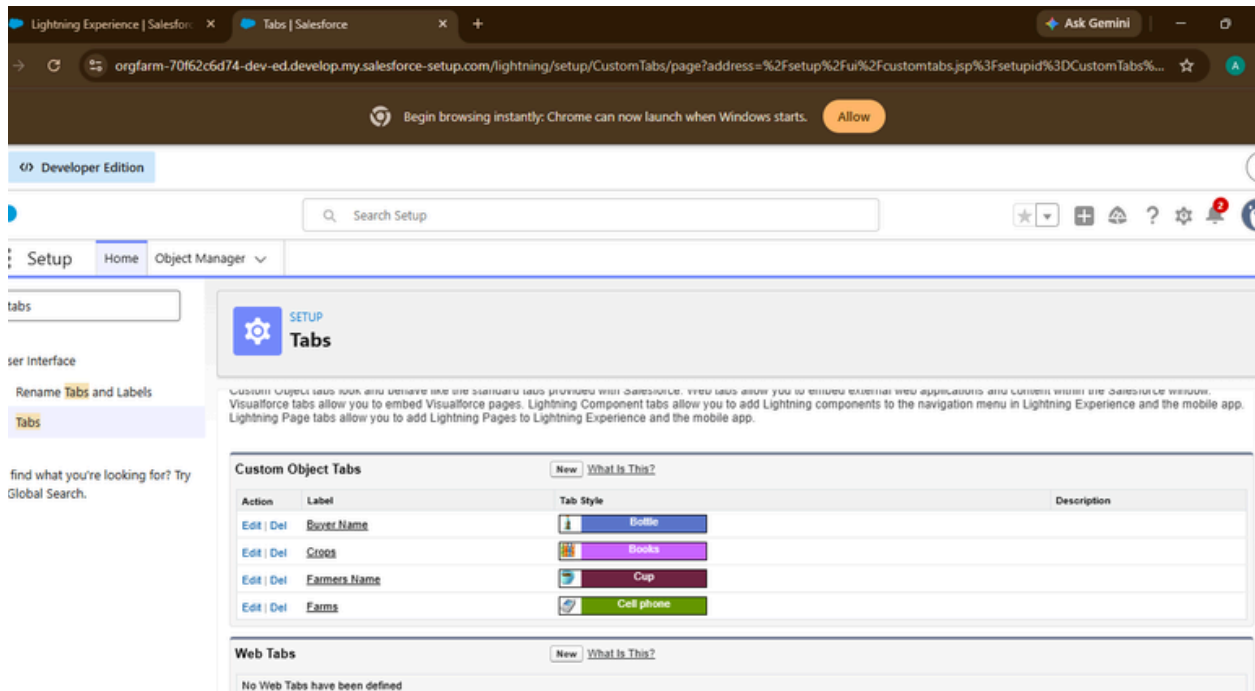
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

#### **Activity 1:**

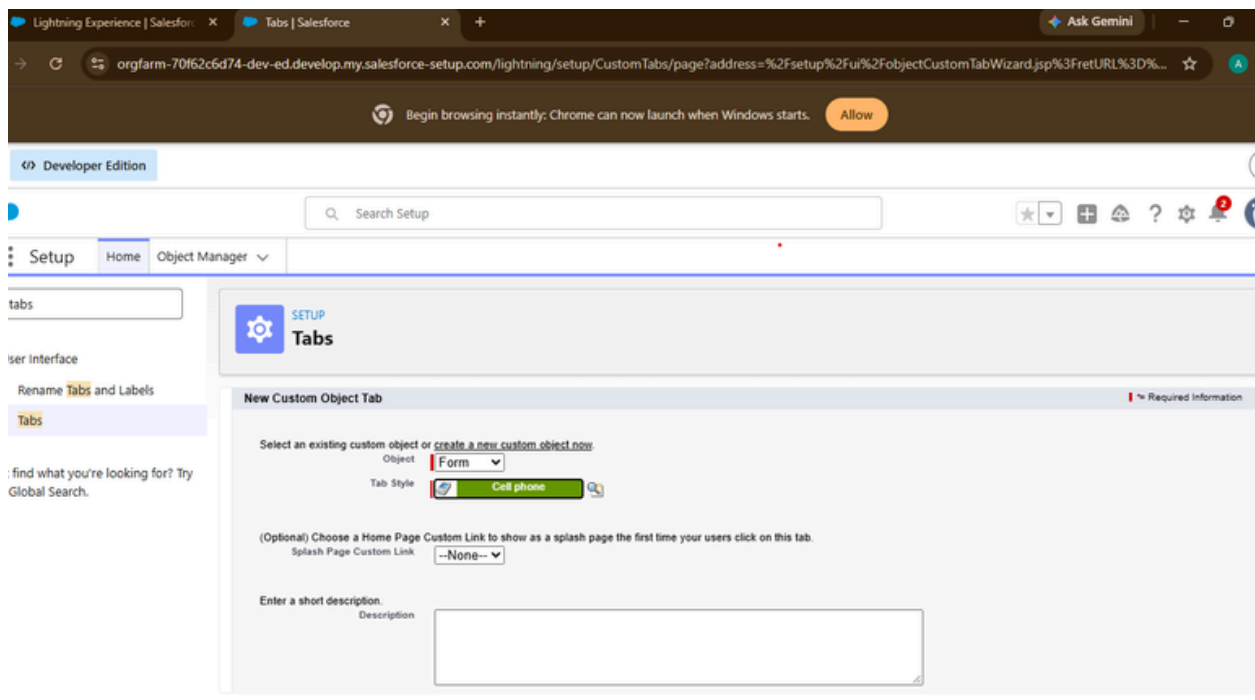
##### **Creation of Farm tab:**

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



2. Under custom object tabs, click New.



3. For Object, select Farm.

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

## **Activity 2:**

### **Creation of Crop tab:**

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

## **Activity 3:**

### **Creation of Farmer tab:**

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

## **Activity 4:**

### **Creation of Buyer tab:**

Now create a custom tab.

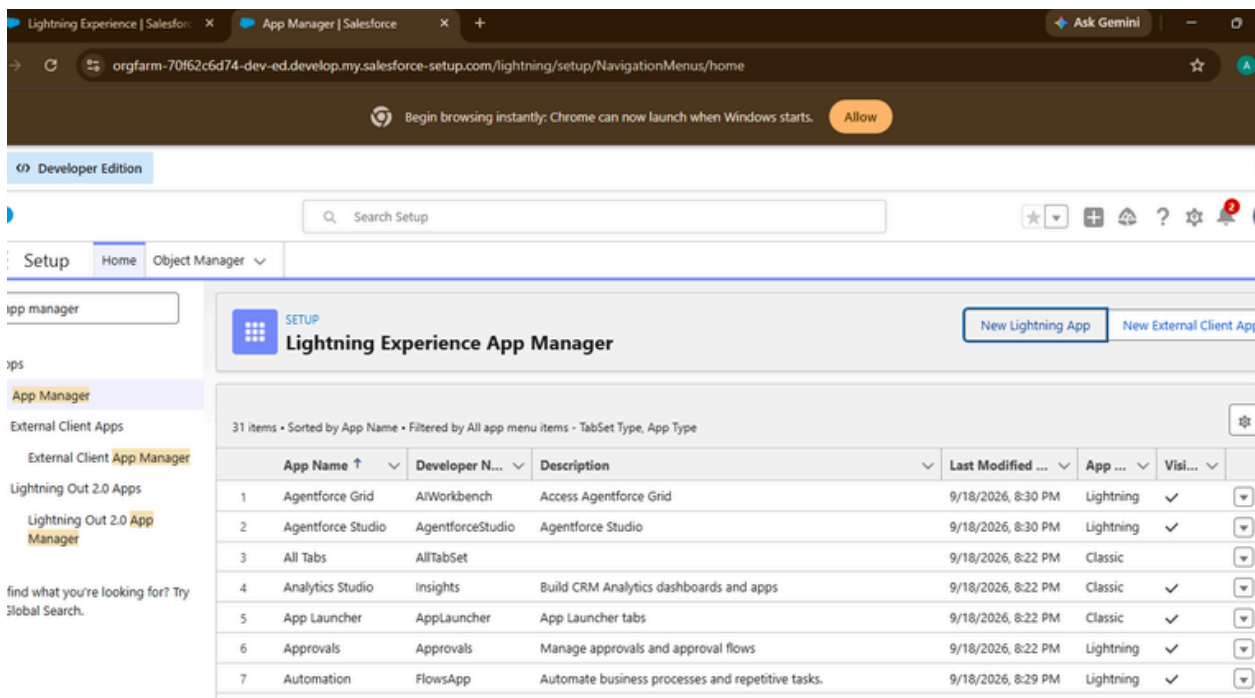
1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

## **Milestone 4 - Lightning App:**

What is an App?

Apps in Salesforce area group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

## Activity 1: Create the Farm Management System app



- From Setup, enter App Manager in the Quick Find and select App Manager.

- Click New Lightning App. Farm Management System as the App Name, then click Next

- Under App Options, leave the default selections and click Next.

- Under Utility Items, leave as is and click Next.

- From Available Items, select Farm, Crop, Farmer, Buyer

Reports, and Dashboards and move them to Selected Items. Click Next. ● From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.

- To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

Note:

1. App Launcher-Displays available apps.

2. App Name-Displays the current selected app.

orgfarm-70f62c6d74-dev-ed.develop.my.salesforce-setup.com/lightning/setup/NavigationMenus/home

Begin browsing instantly: Chrome can now launch when Windows starts. [Allow](#)

Developer Edition

### New Lightning App

#### App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

##### App Details

\*App Name ⓘ

\*Developer Name ⓘ

##### App Branding

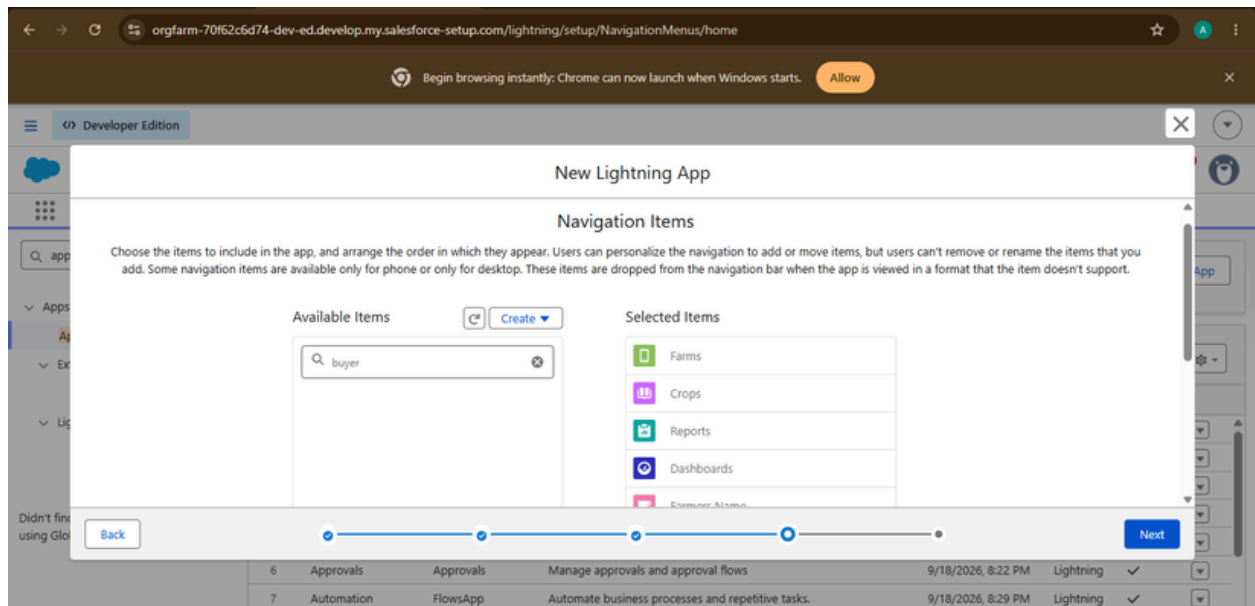
Image ⓘ

Primary Color Hex Value ⓘ

Next

|   |            |           |                                                   |                    |           |   |
|---|------------|-----------|---------------------------------------------------|--------------------|-----------|---|
| 6 | Approvals  | Approvals | Manage approvals and approval flows               | 9/18/2026, 8:22 PM | Lightning | ✓ |
| 7 | Automation | FlowsApp  | Automate business processes and repetitive tasks. | 9/18/2026, 8:29 PM | Lightning | ✓ |

### 3. Navigation menu-Displays the tabs available inside the app.



## Milestone 5 -

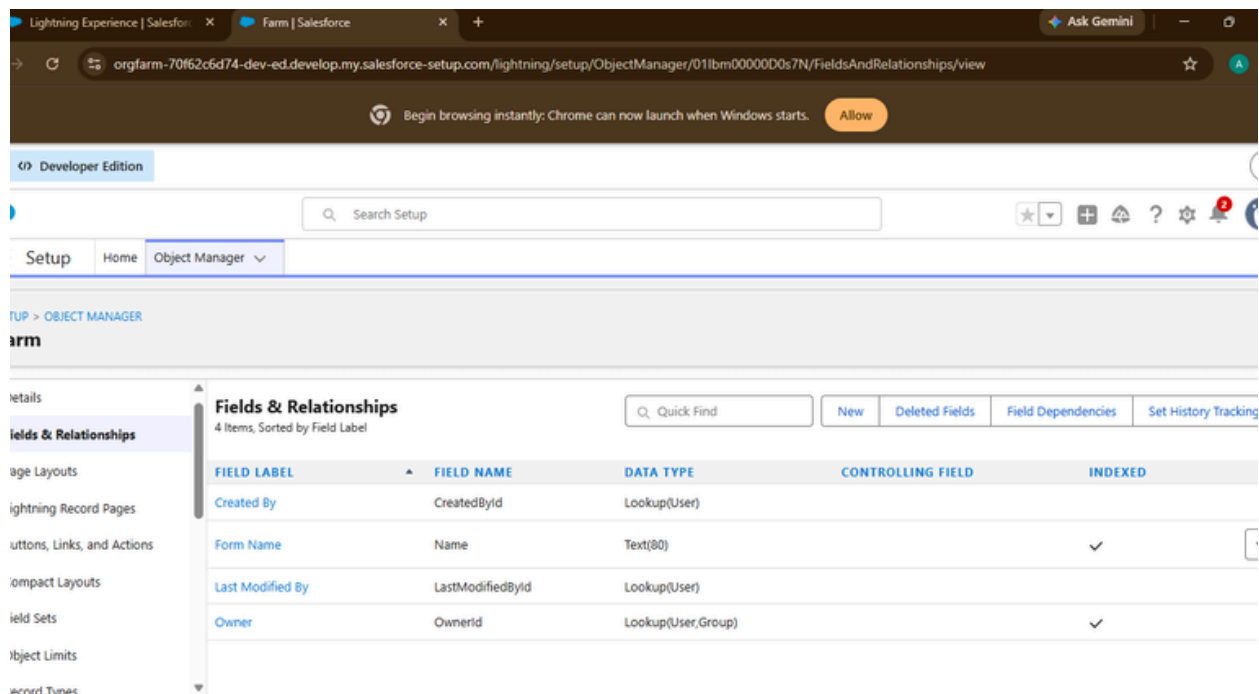
### Fields and Relationship: What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

#### Activity 1:

Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New



The screenshot shows the Salesforce Setup interface for the 'Farm' object. The 'Fields & Relationships' section is active, displaying a table with 4 items, sorted by Field Label. The table columns are FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The items listed are Created By, Form Name, Last Modified By, and Owner.

| FIELD LABEL      | FIELD NAME       | DATA TYPE          | CONTROLLING FIELD | INDEXED |
|------------------|------------------|--------------------|-------------------|---------|
| Created By       | CreatedById      | Lookup(User)       |                   |         |
| Form Name        | Name             | Text(80)           |                   | ✓       |
| Last Modified By | LastModifiedById | Lookup(User)       |                   |         |
| Owner            | OwnerId          | Lookup(User,Group) |                   | ✓       |

## 5. Choose Data type as a picklist, Click next

The screenshot shows the Salesforce Setup interface. The browser address bar displays the URL: `orgfarm-70f62c6d74-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/011bm0000D0s7N/FieldsAndRelationships/new`. The page header includes the 'Developer Edition' label and a 'Search Setup' bar. The main navigation bar shows 'Setup', 'Home', and 'Object Manager'. The left sidebar contains a list of navigation items: 'Fields & Relationships' (highlighted), 'Page Layouts', 'Lightning Record Pages', 'Buttons, Links, and Actions', 'Impact Layouts', 'Id Sets', and 'Object Limits'. The main content area displays a list of data types with radio buttons: Email, Geolocation, Number, Percent, Phone, Picklist (selected), Picklist (Multi-Select), Text, Text Area, and Text Area (Long). To the right of the selected 'Picklist' option, a description is provided: 'Allows users to select a value from a list you define.'

6. Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)
7. Next, Next Click Save.

The screenshot shows the Salesforce Lightning Experience interface for creating a new custom field. The browser address bar indicates the URL: `orgfarm-70f62c6d74-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/011bm00000D0s7N/FieldsAndRelationships/new`. The page title is "New Custom Field" for the "Farm" object. The left sidebar contains navigation links: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, and Object Limits. The main content area is titled "Step 2. Enter the details" and "Step 2 of 4". It features a "Field Label" input field with the text "Farm type" and an information icon. Below this, the "Values" section has two radio button options: "Use global picklist value set" (selected) and "Enter values, with each value separated by a new line". A dropdown menu below the radio buttons shows "--None--". At the bottom, there is a checkbox labeled "Display values alphabetically, not in the order entered." and the Einstein logo.

Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

- 1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)
- 2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)
- 3.Choose data type as a text,enter field label(Location) 4.Choose data type as text,enter field label (size)
- 5.Create a Master Detail relationship with Farmer Object.

## **Activity 2:**

### **Creation of fields for the Crop objects:**

- 1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save. 2.Select Data type as date ,Enter Field label (Planting Date),then save .
- 3.Again we select data type as a date,Enter field labe (Harvest date),save
- 4.Next We Create Lookup relationship on crop object realted to Farm Object.
- 5.We select adata type as currency,Enter field Label (Actual Yield), Next Save it.
- 6.Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.
- 7.Now We create Formula Field For That Select data Type As a Formula,Click Next.

Lightning Experience | Salesforce

Crop | Salesforce

SkillWallet

Ask Gemini

orgfarm-70f62c6d74-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01lbm00000D0sP7/FieldsAndRelationships/new

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

Developer Edition

Setup

Home

Object Manager

Setup > OBJECT MANAGER

Crop

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Specify the type of information that the custom field will contain.

Data Type

☐ None Selected

Select one of the data types below.

☐ Auto Number

A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

☒ Formula

A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.

☐ Roll-Up Summary

A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.

☐ Lookup Relationship

Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

☐ Master-Detail Relationship

Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where:

- The relationship field is required on all detail records.

Next Cancel

8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.

The screenshot shows the Salesforce 'Object Manager' interface for a 'Crop' object. The browser address bar indicates the URL: `orgfarm-70f62c6d74-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01lbm00000D0sP7/FieldsAndRelationships/new`. The page is titled 'Developer Edition' and 'Setup'. The left sidebar shows the navigation menu with 'Fields & Relationships' selected. The main content area is 'Step 3. Enter formula' for the 'Crop' object. It includes a 'Previous', 'Next', and 'Cancel' button bar. Below the header, there is an example formula: `Gross Margin = Amount - Cost`. The 'Simple Formula' tab is active, showing a 'Select Field Type' dropdown set to 'Crop' and an 'Insert Field' dropdown set to '-- Insert Merge Field --'. A 'Use Formula Assistant' button and an 'Insert Operator' dropdown are also present. The formula text area contains: `Expected Yield per Acre) (Currency) = Actual Yield / Total acre Harvested`. A 'Quick Tips' box on the right lists 'Getting Started' and 'Operators & Functions'. The Einstein logo is visible in the bottom right corner.

### **Activity 3:**

#### **Creation of fields for the Farmer objects:**

1. Select Data type Phone, Enter field label (Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label (State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

#### **Creation of field dependency :**

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer object, select Fields & Relationships, On Right side We have Field Dependency option, select that option.
2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.
5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.
6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.

### **Activity 3:**

#### **Creation of fields for the Farmer objects:**

1. Select Data type Phone, Enter field label (Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label (State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

#### **Creation of field dependency :**

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer object, select Fields & Relationships, On Right side We have Field Dependency option, select that option.
2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.
5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.
6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.

Search Setup

Setup Home Object Manager

SETUP > OBJECT MANAGER  
**Farmer**

Details

**Fields & Relationships**

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Object Access

### New Field Dependency

Create a dependent relationship that causes the values in a picklist or multi-select picklist to be dynamically filtered based on the value selected by the user in another field.

- The field that drives filtering is called the "controlling field." Standard and custom checkboxes and picklists with at least one and less than 300 values can be controlling fields.
- The field that has its values filtered is called the "dependent field." Custom picklists and multi-select picklists can be dependent fields.

**Step 1.** Select a controlling field and a dependent field. Click Continue when finished.

**Step 2.** On the following page, edit the filter rules that control the values that appear in the dependent field for each value in the controlling field.

|                   |      |          |        |
|-------------------|------|----------|--------|
| Controlling Field | None | Continue | Cancel |
| Dependent Field   | None | Continue | Cancel |

Developer Edition

Search Setup

Setup Home Object Manager

Quick Find

Home

Force Go

Cloud Reports

Force Setup Assistant

Force Assistant

Updates

Force Mobile App

Usage

Exchange Offers

Cloud Everywhere

ADMINISTRATION

SETUP

Click button to include or exclude selected values from the dependent picklist:

Include Values Exclude Values

| state:    | Telangana       | Andhra Pradesh  | Tamil Nadu      |
|-----------|-----------------|-----------------|-----------------|
| District: | Vellore         | Vellore         | Vellore         |
|           | thiruvannamalai | thiruvannamalai | thiruvannamalai |
|           | madurai         | madurai         | madurai         |

Showing Columns: 1 - 3 (of 3) < Previous | Next > | View All | Go to

Click button to include or exclude selected values from the dependent picklist:

Include Values Exclude Values

Save Cancel Preview

7. Click Close to close the preview window, Click Save.

#### **Activity 4:**

##### **Creation of fields for the Buyer objects:**

1. Select Data type as a Phone, Enter a Field Label(Contact), Save.
2. Select Data type as an Email, Enter Field Label(Email), Next save.
3. Select Data type Text area, Enter Field Label(Address), Save it.
4. Select Data type Picklist, Field Label(Buyer Type), Enter values (Wholesale, Retail), Click Save.

#### **Milestone 6- Profile:**

##### **What is a profile?**

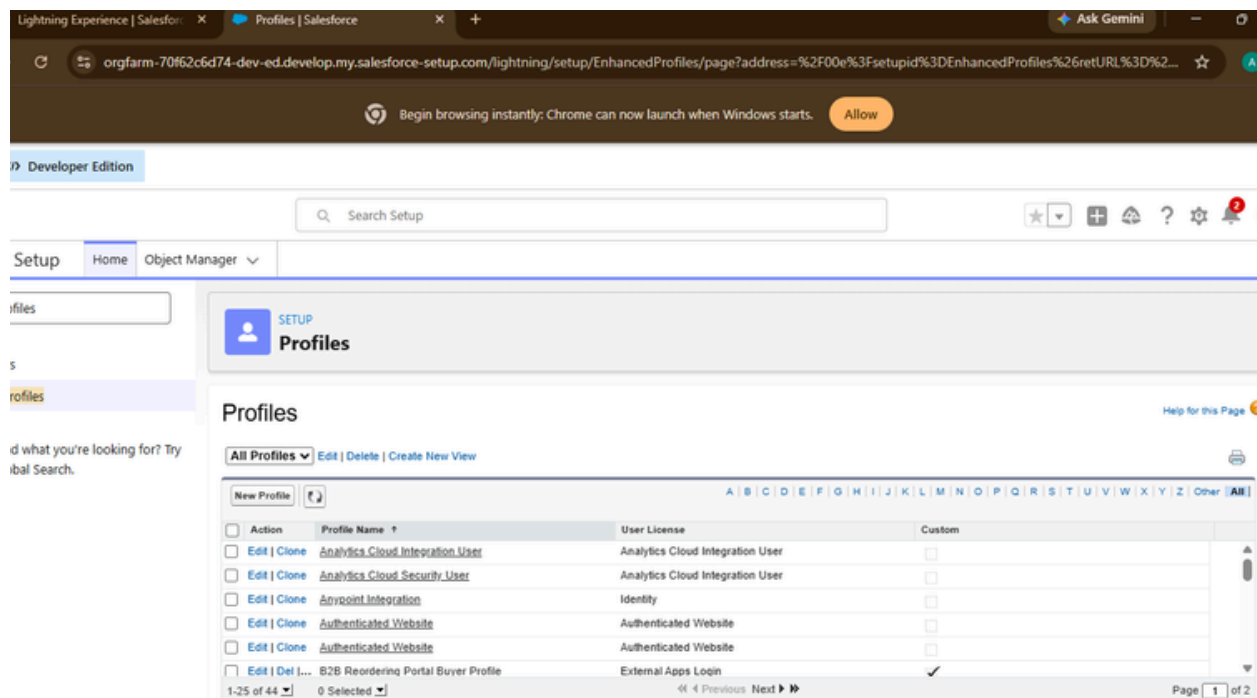
A profile is a group/collection of settings and permissions that define what a user can do in

salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.  
A profile can be assigned to many users, but user can be assigned single profile at a time.

## Activity 1:

### Create a customprofile:

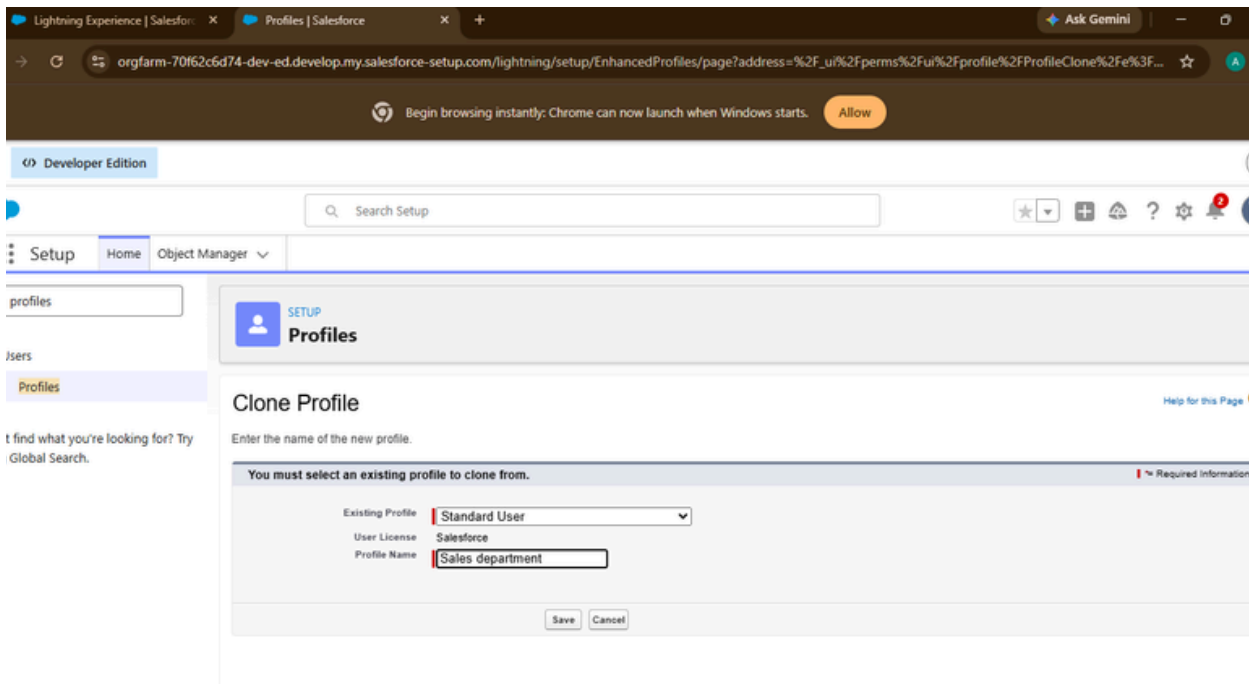
1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.



The screenshot shows the Salesforce Setup interface. The left sidebar contains navigation links for Setup, Home, and Object Manager. The main content area is titled 'Profiles' and includes a search bar and a 'New Profile' button. Below this is a table listing various profiles. The table has four columns: Action, Profile Name, User License, and Custom. The 'Custom' column contains checkboxes for each profile. The 'External Apps Login' profile has a checkmark in the 'Custom' column.

| Action                                    | Profile Name                        | User License                     | Custom                              |
|-------------------------------------------|-------------------------------------|----------------------------------|-------------------------------------|
| <input type="checkbox"/> Edit   Clone     | Analytics Cloud Integration User    | Analytics Cloud Integration User | <input type="checkbox"/>            |
| <input type="checkbox"/> Edit   Clone     | Analytics Cloud Security User       | Analytics Cloud Integration User | <input type="checkbox"/>            |
| <input type="checkbox"/> Edit   Clone     | Appoint Integration                 | Identity                         | <input type="checkbox"/>            |
| <input type="checkbox"/> Edit   Clone     | Authenticated Website               | Authenticated Website            | <input type="checkbox"/>            |
| <input type="checkbox"/> Edit   Clone     | Authenticated Website               | Authenticated Website            | <input type="checkbox"/>            |
| <input type="checkbox"/> Edit   Del   ... | B2B Reordering Portal Buyer Profile | External Apps Login              | <input checked="" type="checkbox"/> |

Page 1 of 2



5. Click save.

## Activity 2:

### Create a customprofile-2:

Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box
7. Select profiles (Standard user).
8. Click clone.

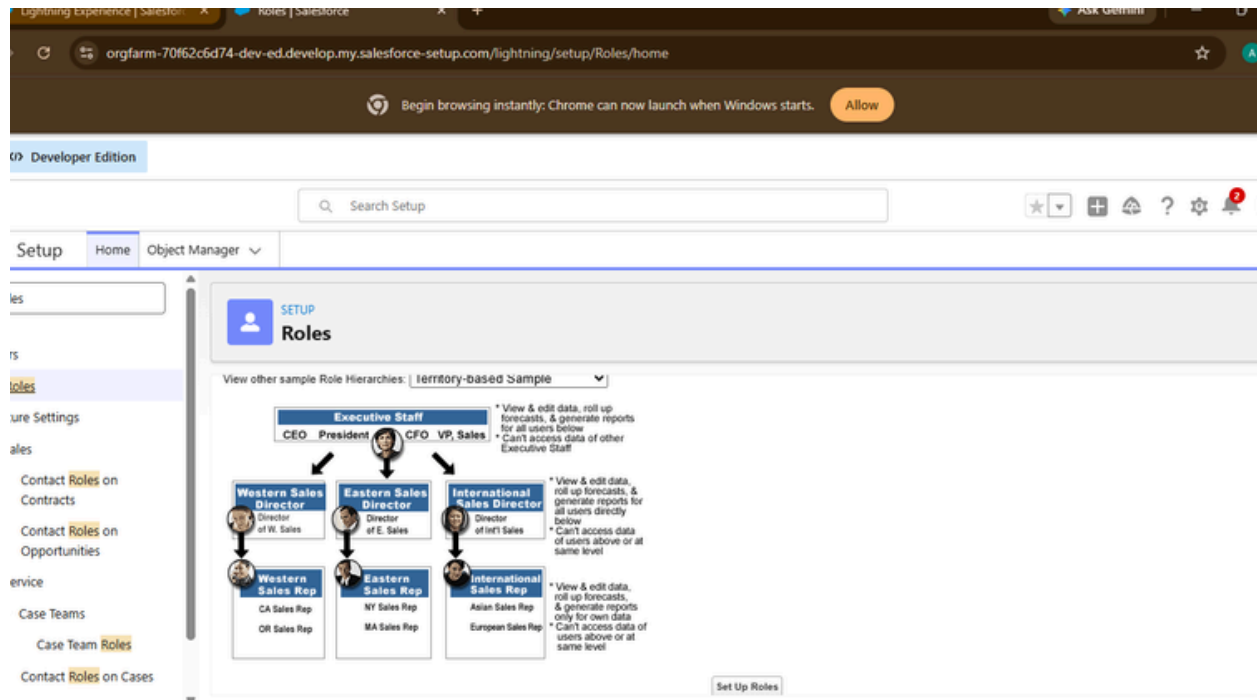
## Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

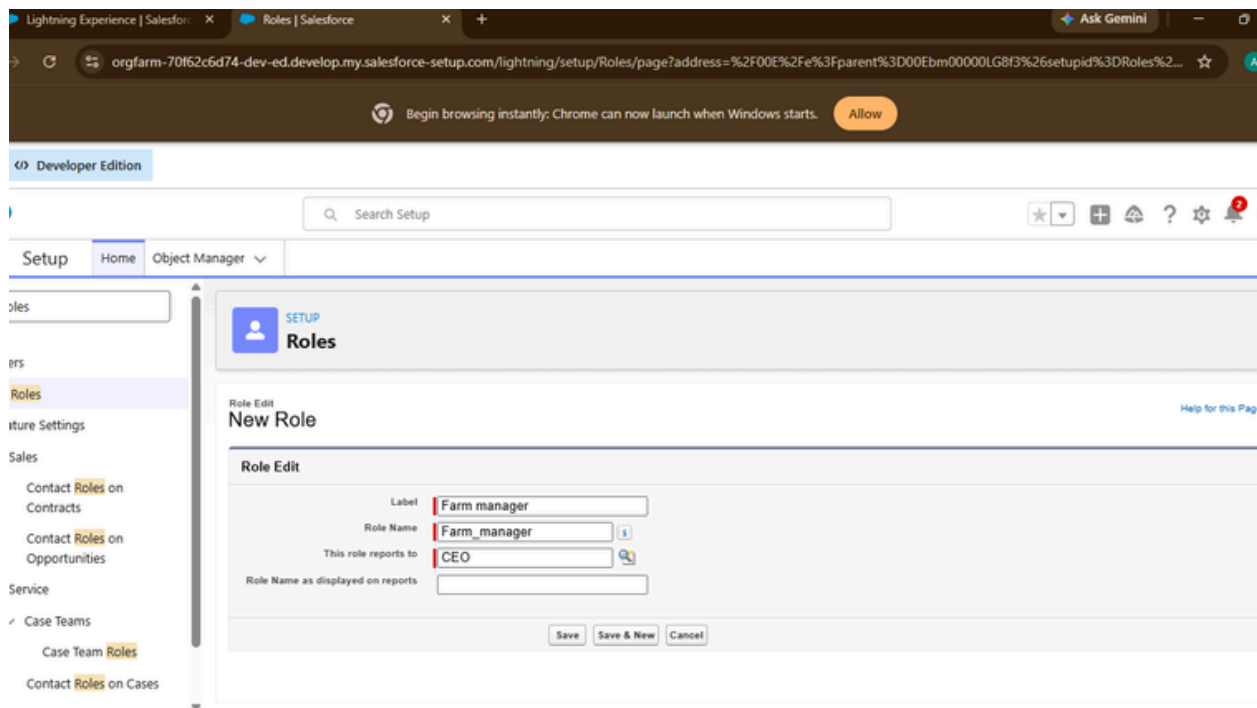
## Activity-1

## Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option



3. Below the CEO click on add role and enter the label name as a " Farm Manager " and role name will be Automatically populated and click on save.



## Milestone 8-User :

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in

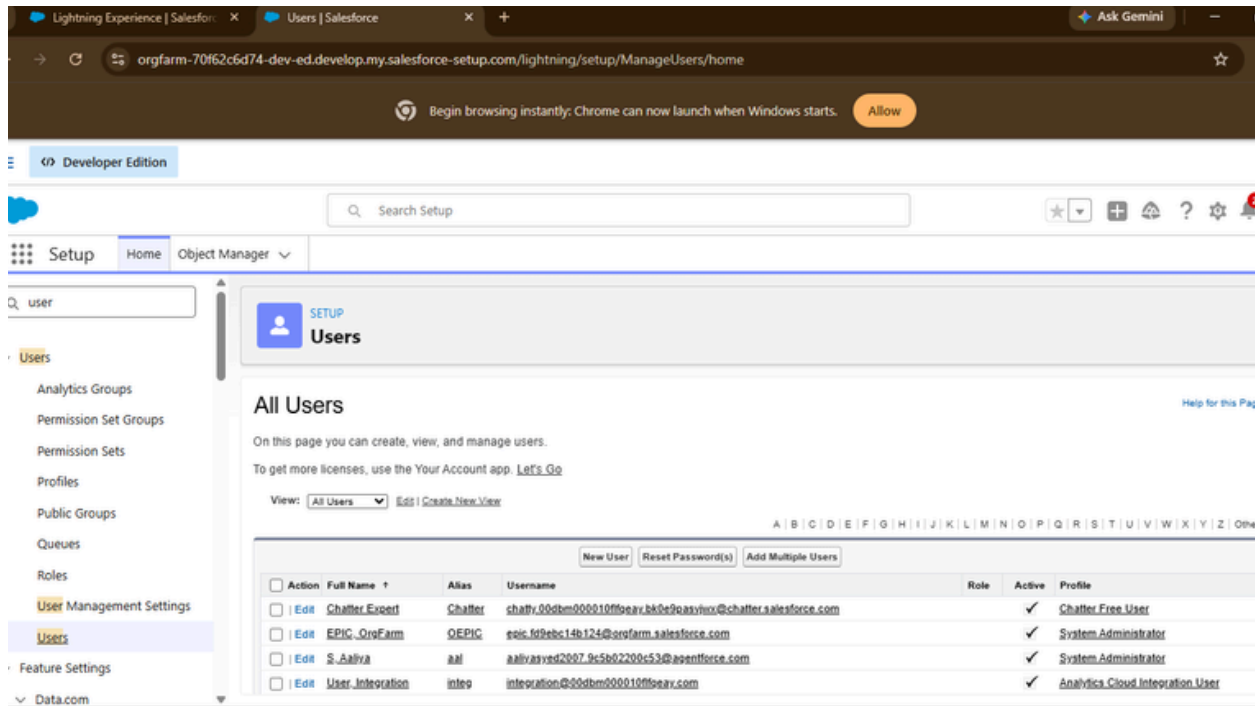
Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

**NOTE-** As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

## Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.



3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

## Activity 2:

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

## Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

- Owner-based Sharing Rules
- Criteria-based Sharing Rules

## Activity 1:

### Create a sharing rule

- 1) Goto Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with the section select roles and in that select Farm Manager.
- 8) And in the section of select the level of access for the users give the access Read/Write.
- 9) And save the rule.

The screenshot shows the Salesforce 'Sharing Settings' page in a web browser. The page is titled 'Sharing Settings' and includes a 'SETUP' icon. Below the title, there is a note: 'Use sharing rules to make additional exposures to your organization-wide sharing settings for selected sets of users. Note: "Roles and subordinates" includes all users in a role, and the roles below that role. You can use sharing rules only to grant wider access to data, not to restrict access.'

The page is divided into three steps:

- Step 1: Rule Name** (Required Information)
  - Label:
  - Rule Name:
  - Description:
- Step 2: Select your rule type**
  - Rule Type: ☐ Based on record owner ☒ Based on criteria
- Step 3: Select which records to be shared**

## Activity 2:

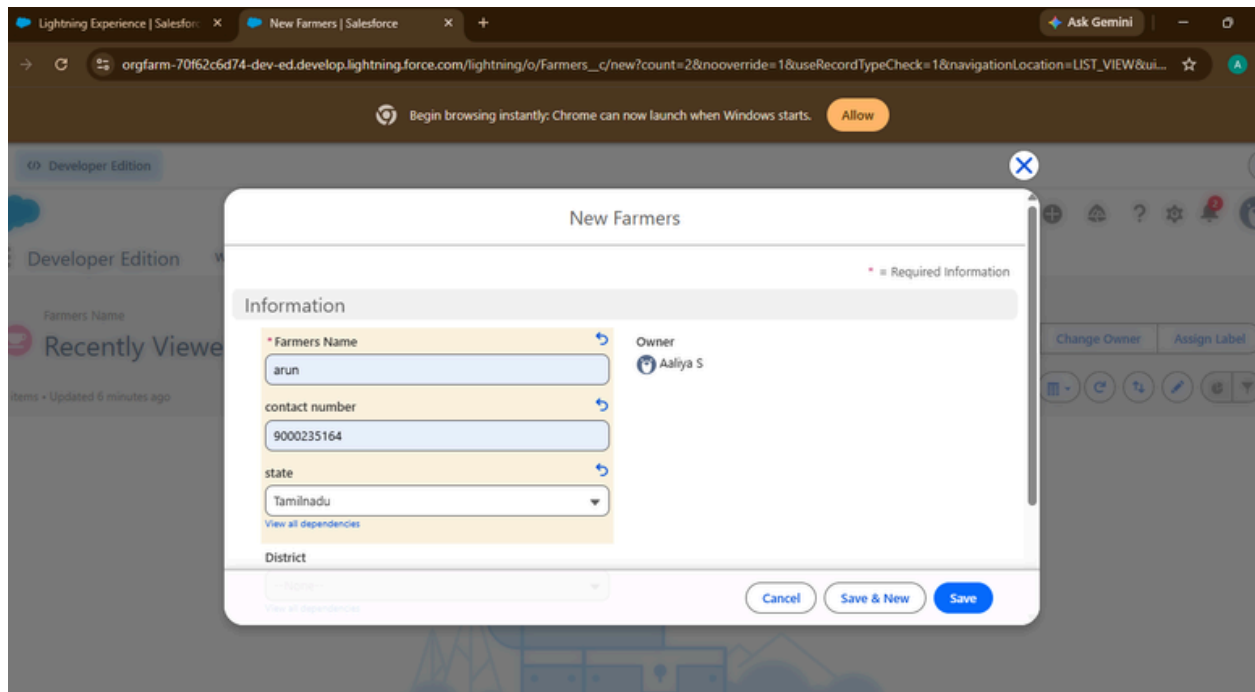
Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

## Milestone-10: User Adoption

### Activity 1: Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.



The screenshot shows a web browser window with the Salesforce Lightning Experience interface. The browser's address bar displays the URL: `orgfarm-70f62c6d74-dev-ed.develop.lightning.force.com/lightning/o/Farmers__c/new?count=2&nooverride=1&useRecordTypeCheck=1&navigationLocation=LIST_VIEW&ui...`. A notification bar at the top states, "Begin browsing instantly: Chrome can now launch when Windows starts." with an "Allow" button. The main content area features a "New Farmers" modal form. The form has a title bar with a close button (X) and a "Required Information" indicator. The form is divided into two sections: "Information" and "District". The "Information" section contains four fields: "Farmers Name" (text input with value "arun"), "contact number" (text input with value "9000235164"), "state" (dropdown menu with value "Tamilnadu"), and "Owner" (lookup field with value "Aaliya S"). The "District" section contains a dropdown menu with the value "None". At the bottom of the form are three buttons: "Cancel", "Save & New", and "Save".

## Milestone 11-Reports:

### What are Reports?

A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

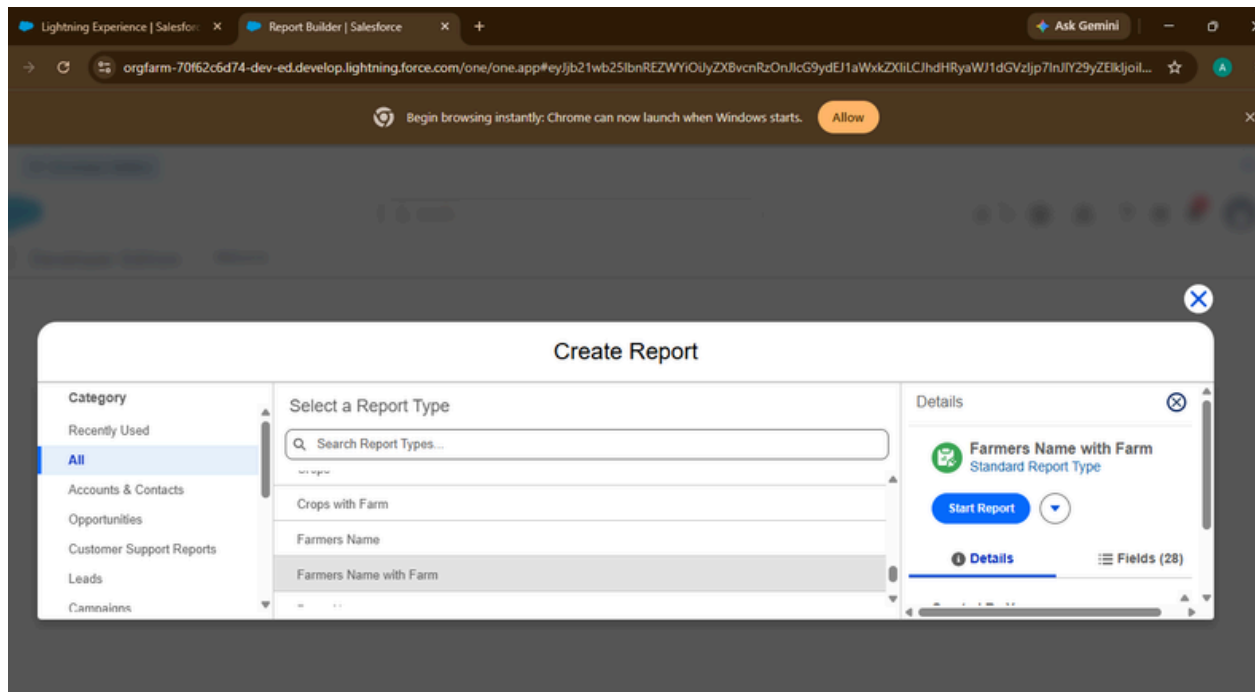
### Activity 1:

#### Create a report:

Create a report that displays rating of the account and which has type and account name.

1) Click on app launcher search for reports.

2) Click on the new report and select the category has Farmer with Farms.

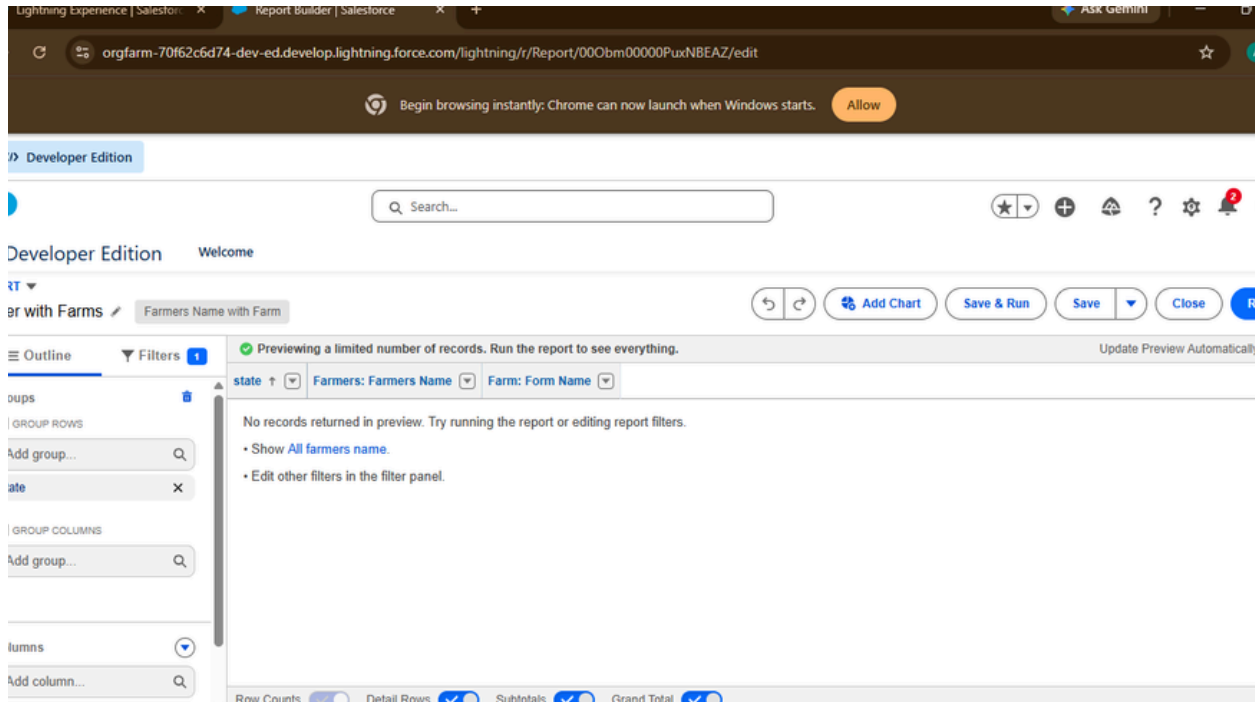


3) In the details section select the option start report.

4) In Outline Click on Column Select Which Field you want to display in the Report.

5) Click Group, Select Which Fields You Want to Group.

6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.



## Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

## Milestone 12-Dashboard:

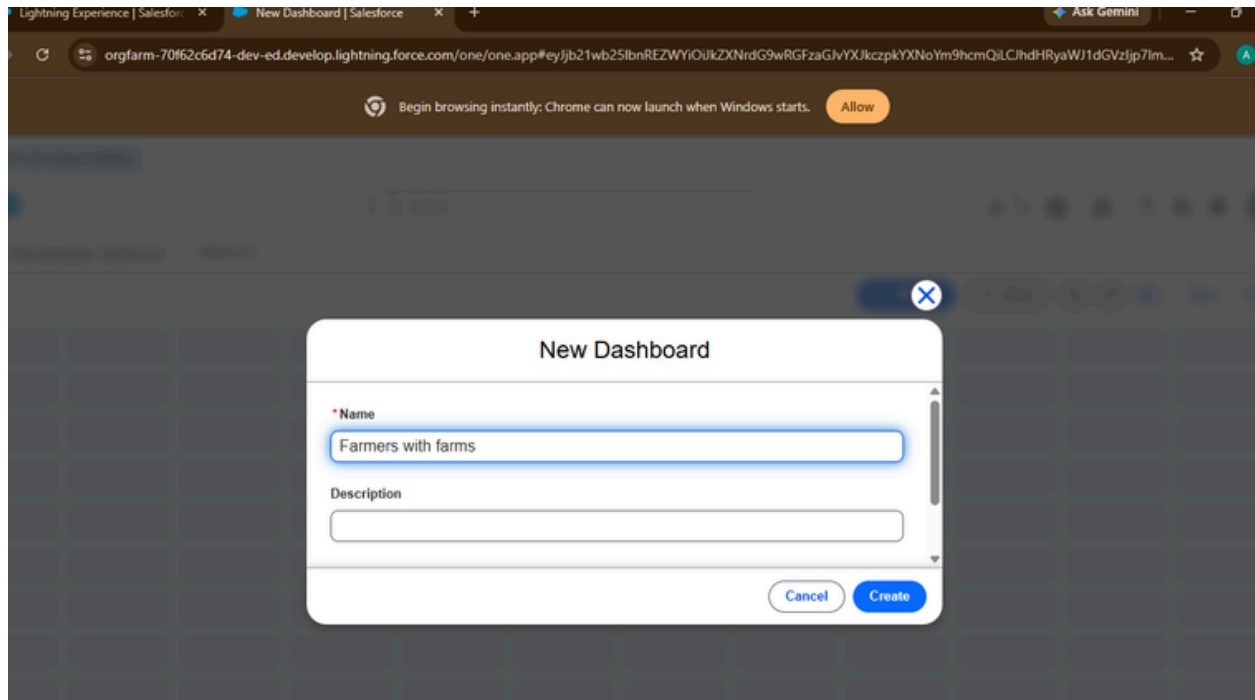
Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

## Activity 1:

### Create a Dashboard:

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.

7. Click Save and then Done.



### **Milestone-13: View Report and Dashboard:**

#### **Activity1:Report**

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records

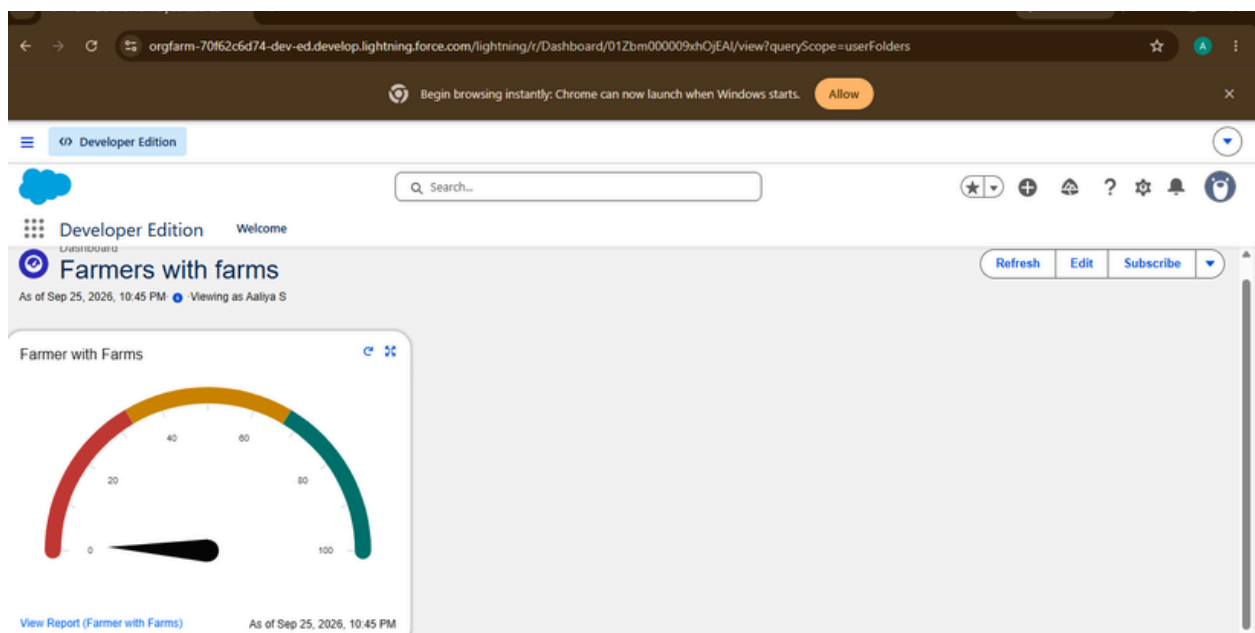
Total Records  
2

| state                                   | Farmers: Farmers Name | Farm: Form Name |
|-----------------------------------------|-----------------------|-----------------|
| <input type="checkbox"/> Telangana, (1) | rajesh                | paddy farm      |
| Subtotal                                |                       |                 |
| <input type="checkbox"/> Taminadu (1)   | arun                  | arun farm       |
| Subtotal                                |                       |                 |
| Total (2)                               |                       |                 |

Row Counts Detail Rows Subtotals Grand Total

## Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



## **7. FUNCTIONAL & PERFORMANCE TESTING**

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

## **8. ADVANTAGES & DISADVANTAGES**

### **Advantages**

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

### **Disadvantages**

- Requires internet connection
- Salesforce license dependency
- Limited offline access

## **9. Conclusion:**

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

## **10. FUTURE SCOPE**

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

## **11. APPENDIX**

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

**Thank You**